

Debdip Bandyopadhyay

Senior SDET | AI Quality Engineering Lead

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PROFESSIONAL SUMMARY

Senior Software Development Engineer in Test (SDET) with **11+ years of experience** architecting enterprise-grade automation frameworks, AI quality evaluation pipelines, and quality engineering strategies across **Generative AI platforms, Aerospace, and Enterprise Systems**. Core member of the GenAI CoE team with proven ownership of end-to-end testing across cross-domain engineering teams. Deep expertise in **LLM Evaluation (DeepEval, Semantic Similarity, Faithfulness)**, Python test automation (Playwright, Pytest), and CI/CD quality gates. Master's degree from IIT Jodhpur (**School of AI & Data Science**) with independent published research in LLM provenance, model DNA attribution, and cloze scaling laws (CERN Zenodo DOIs: 10.5281/zenodo.22158286 & 10.5281/zenodo.22158483).

TECHNICAL SKILLS

- **AI Quality & LLM Evaluation:** LLM Testing, RAG Validation, DeepEval, Semantic Similarity Assessment, Prompt & Non-Deterministic Output Evaluation, Hallucination & Faithfulness Scoring, NVIDIA NIM, PyTorch.
- **Automation Frameworks:** Python, Playwright, Pytest, Pytest-BDD, Cypress, API Testing (Requests, Postman), Page Object Model (POM), Parallel Execution (pytest-xdist), Allure Reporting, Tricentis Tosca, UFT.
- **DevOps & CI/CD:** Azure DevOps Pipelines, Docker, Git, Linux, Poetry.
- **Enterprise & Methodologies:** Oracle EBS (AP, AR, PO, CE, FA), Jira, Agile / Scrum, Zero-Trust Quality Gates.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Apr 2021 – Present

Senior Test Automation Engineer / Senior SDET – GenAI CoE Team

Kolkata, India

- **GenAI CoE Quality Ownership:** Served as a core member of the GenAI Center of Excellence (CoE) team, taking complete end-to-end testing ownership and collaborating across cross-domain developer teams (NLP engineers, full-stack developers, cloud architects) to deliver production-ready AI solutions.
- **AI Evaluation Pipelines:** Implemented automated AI evaluation suites for LLM and RAG conversational applications using DeepEval, validating non-deterministic outputs across correctness, faithfulness, context relevancy, and semantic similarity.
- **Framework Architecture:** Architected an enterprise-grade Playwright + Python + Pytest-BDD automation framework adopting modular Page Object Model (POM) design patterns.
- **CI/CD Quality Gates:** Integrated automated test execution into Azure DevOps CI/CD pipelines, enabling continuous quality validation on every build release.
- **Performance & Speed:** Reduced regression execution cycle times by ~40% through parallel execution (pytest-xdist), headless browser optimization, and selective test sharding.
- **Enterprise Automation:** Automated end-to-end business workflows for Oracle EBS modules (AP, AR, PO, CE, FA) utilizing UFT and Tricentis Tosca.
- **Cross-Domain Leadership:** Mentored 10+ automation engineers and partnered with Architects, Data Scientists, and Product Owners to define enterprise-wide quality benchmarks.

The Boeing Company

Oct 2018 – Apr 2021

Software Engineer – Test Automation

Bengaluru, India

- **Safety-Critical Testing:** Engineered test automation solutions for safety-critical aerospace software operating within strictly regulated engineering standards.
- **Regression Optimization:** Maintained and optimized enterprise regression suites of 2,000+ automated test cases across multi-phase release cycles.
- **Mentorship & Quality Standards:** Mentored junior test engineers on coding standards and code reviews.
- **Recognition:** Awarded Boeing Pride Award and Star Award (2019) for outstanding contributions to testing critical applications in short sprints.

PROFESSIONAL EXPERIENCE (CONTINUED)

Accenture LLP <i>Test Automation Engineer</i>	Oct 2014 – Oct 2018 <i>Kolkata / Noida, India</i>
• Multi-Platform Automation: Developed reusable, data-driven automation frameworks across Web, Desktop, SAP, and Mobile applications. • Suite Automation: Automated end-to-end Integration, System Regression, Smoke, and User Acceptance Testing (UAT) workflows. • Cost Reduction Utility: Designed and deployed an in-house One Click Capture test utility, reducing third-party tooling expenses by ~USD 8,000 annually. • Client Engagement: Collaborated with business stakeholders to translate functional requirements into proof-of-concept (PoC) automation suites.	

PUBLICATIONS & AI RESEARCH PREPRINTS

- **Multi-Scale 4-Pass Sentence Cloze Infilling with Pass-Adaptive Dynamic Gating for Robust AI Text Detection**
[Zero-shot black-box framework achieving **99.58% AUROC on RAID (ACL 2024)**, **99.98% on HC3**, and **0.00% FPR on human literature** (d=4.07, p=7.49e-4) via Kuhn-Munkres S3 bipartite matching and continuous sigmoidal gating]
[CERN Zenodo DOI: 10.5281/zenodo.22158286](#) | [SSRN: 7341078](#) | [Benchmarks: github.com/debdipARVR/cloze_congruence_reproducibility_benchmark_kit](#)
- **ClozeCongruence: Model Fingerprinting and Traceback Attribution in LLMs via Reverse Cloze Resonance and Micro-Stylometric DNA**
[Foundation model provenance traceback achieving **100.00% Top-1 attribution accuracy** (Macro-F1=1.0000) across 6 frontier LLMs (GPT-4o, Claude 3.5, Gemini 3.7, Nemotron-3 120B, GLM-5.2) with +97.23% resonance gap and 12D micro-stylometrics]
[CERN Zenodo DOI: 10.5281/zenodo.22158483](#) | [SSRN: 7368119](#) | [Interactive Demo: huggingface.co/spaces/DebdipCS/cloze-congruence-ai-forensics-hub-v1](#)
- **The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Multi-Model Provenance Traceback**
[Systematic empirical scaling study establishing monotonic prober recovery (12B to 120B) and GLM-5.2 native blank-infilling inductive advantage (+51.16% separation gap, Cohen's d=5.62, p=2.87e-15)]
[SSRN Series: 7368141](#) | [ResearchGate & Preprints.org](#) | [Open Logs: github.com/debdipARVR/cloze_congruence_reproducibility_benchmark_kit](#)

EDUCATION

M.Tech in Augmented Reality / Virtual Reality <i>Indian Institute of Technology Jodhpur (School of Artificial Intelligence & Data Science)</i>	Jul 2022 – Dec 2025 <i>Jodhpur, India</i>
B.Tech in Computer Science & Engineering <i>Greater Noida Institute of Technology (AKTU University)</i>	Apr 2010 – Jun 2014 <i>Greater Noida, India</i>
Higher Secondary (Class XII) – CBSE [PCMB] <i>Kendriya Vidyalaya No. 1, Salt Lake (KV No. 1 SLK)</i>	2010 <i>Kolkata, India</i>

CERTIFICATIONS

- Postman API Testing Path (v12) • ASDA-MIT Certification • Google Crash Course on Python (Coursera)